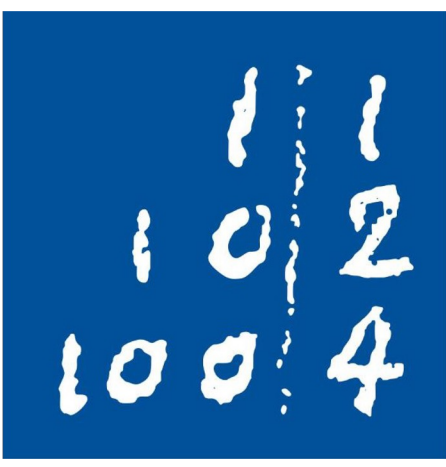


Memory Reach: GRU vs MLP under PPO on MiniGrid Memory

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Code, studies and
winning checkpoints



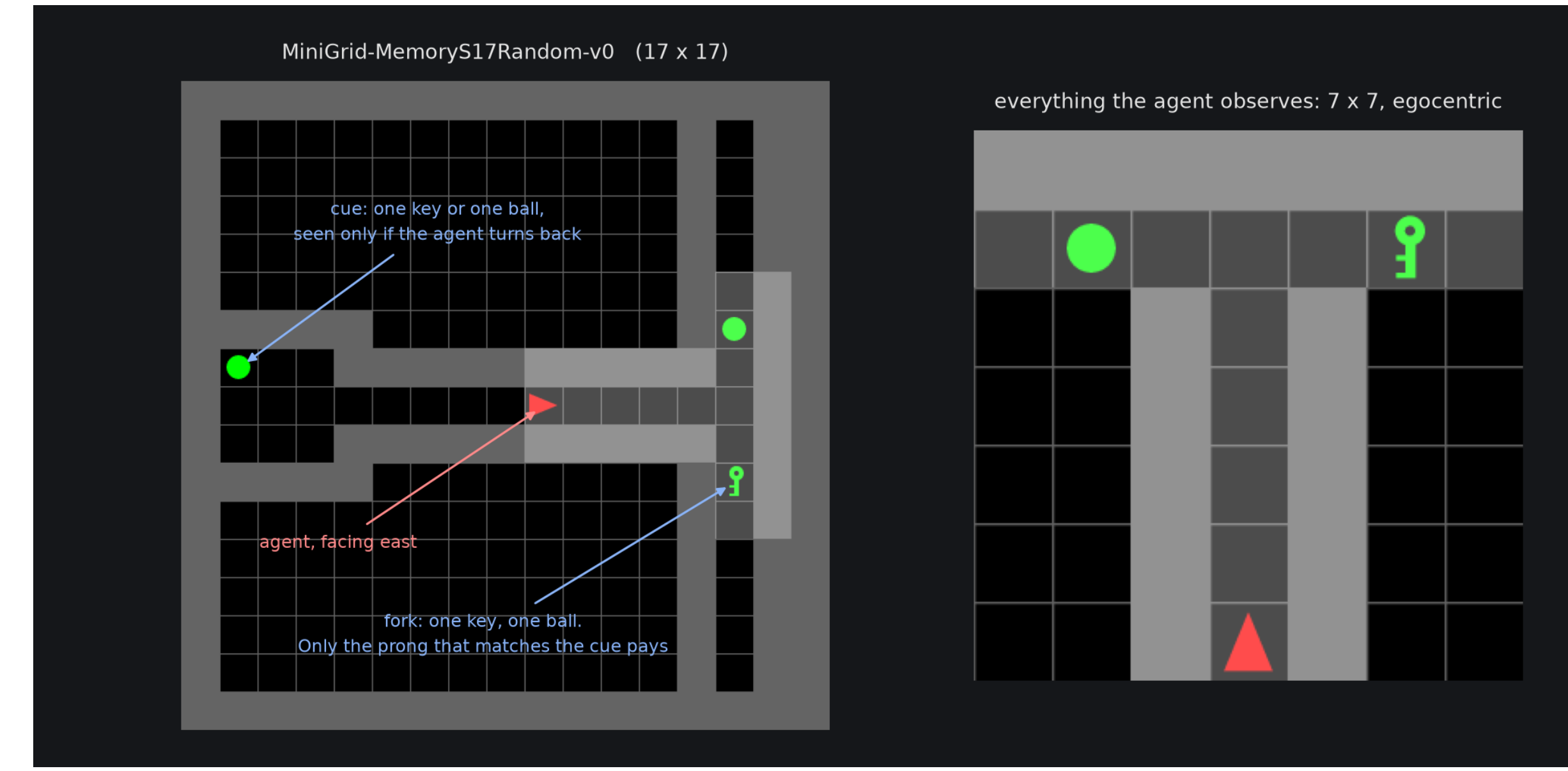
1 Motivation

- **POMDP**: only a 7×7 view is seen
 - **Memory** must live in a state h
 - **Full BPTT** costs memory $\propto$ length
 - **Truncation** cuts gradients to L
- Core question:** how far back must the gradient reach, and what does that reach cost?

250 runs · $\approx$ 250 GPU-hours · RTX 3060 12 GB · 16 CPUs

2 Problem Setting

- The environment
- MiniGrid-MemoryS17Random-v0
 - Cue (key or ball) in the west room
 - Agent spawns facing away from it
 - T-junction at the far east end
 - Only the matching prong pays
 - Reward $1 - 0.9 \cdot \text{steps} / \text{max_steps}$
- The two encoders
- **MLP**: linear layers with ReLU, memory less
 - **GRU**: carries h , with memory



Left: the maze. Right: everything the agent sees.

5 Key Insights

- 1 **Carrying h forward is not what matters.** Every GRU arm does it; only the reach of the gradient differs.
- 2 **What matters is the cue→decision gap.** Not the episode length. ≤ 8 steps in 62 % of mazes, ≤ 4 in 0%.
- 3 **The reach costs memory, not time.** Full BPTT is the fastest GRU arm (1.3 h); it pays in VRAM.

Recommendation. Full BPTT; $L = 8$ on a smaller GPU.

3 Approach

Network

actor-critic skeleton

the encoder is the only thing that changes between arms

GRU encoder, unrolled

$r_t = \sigma(W_r x_t + U_r h_{t-1})$

$z_t = \sigma(W_z x_t + U_z h_{t-1})$

$n_t = \tanh(W_n x_t + r_t \odot U_n h_{t-1})$

$h_t = (1 - z_t) \odot n_t + z_t \odot h_{t-1}$

probability ratio

clipped policy loss

value loss

what is backpropagated

PPO objective

$$\rho_t = \pi_\theta(a_t | s_t) / \pi_{\theta_{old}}(a_t | s_t)$$
$$\mathcal{L}^\pi = -\mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t)]$$
$$\mathcal{L}^V = \mathbb{E}_t[(V_\theta(s_t) - \hat{R}_t)^2]$$
$$\mathcal{L} = \mathcal{L}^\pi + c_v \mathcal{L}^V - c_e \mathcal{H}[\pi_\theta]$$

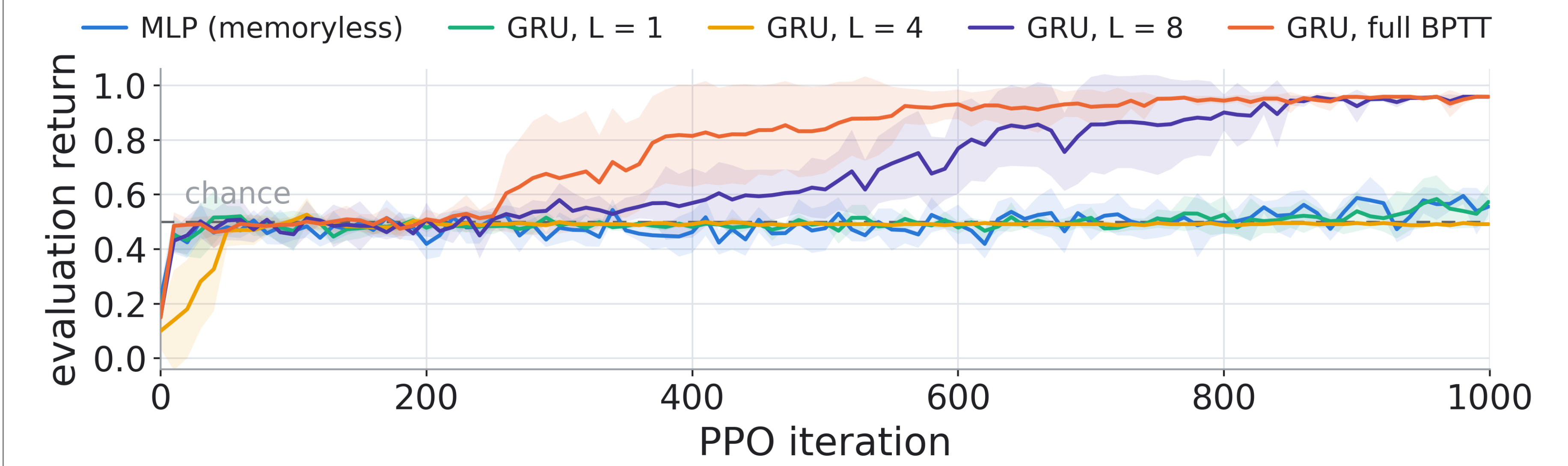
Hyperparameter Optimization

- **Surrogate model:** Optuna TPE sampler, seeded at 42
- **Budget:** 50 trials per arm $\times$ 5 arms = 250 training runs
- **Seeds per trial:** 5, each 1000 iterations
- **Objective:** mean – std of return over the 5 seeds

4 Results

arm	return	success rate	steps	minibatch	time
MLP (memoryless)	0.556 ± 0.029	0.592 ± 0.027	24.3 ± 2.3	1024	1.3 h
GRU, $L = 1$	0.573 ± 0.064	0.588 ± 0.068	9.8 ± 4.1	4096	2.6 h
GRU, $L = 4$	0.492 ± 0.001	0.500 ± 0.000	7.0 ± 0.0	4096	1.6 h
GRU, $L = 8$	0.958 ± 0.004	1.000 ± 0.000	16.8 ± 1.4	4096	2.0 h
GRU, full BPTT	0.958 ± 0.002	1.000 ± 0.000	16.8 ± 0.9	512	1.3 h

Winning trial per study; mean $\pm$ std across the five seeds.



6 Limitations & Future Work

- One task, one architecture, one algorithm
- **GRU is slow:** the recurrence runs step by step
- Next: **LSTM** for a bigger environment
- Next: **transformer** for a faster training time

References. Hausknecht & Stone 2015 · Kapturowski et al. 2019 (R2D2) · Ni et al. 2022 · Schulman et al. 2017 (PPO) · Chevalier-Boisvert et al. 2023 (MiniGrid)